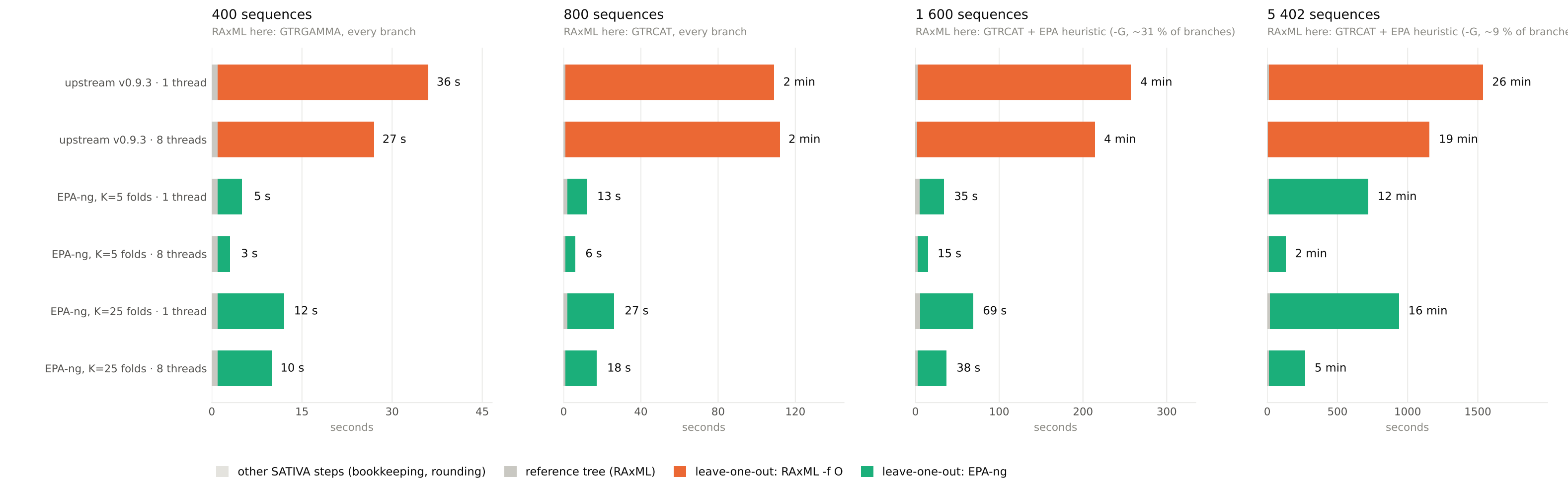


SATIVA with EPA-ng: runtime and agreement

Nested subsets of one ITS alignment. Every run applies the same SATIVA decision rule and builds its reference tree with RAxML 8.2.3 (grey). What changes is the placement engine: the leave-one-out (coloured) and, inside "other steps", the final confirmation pass.

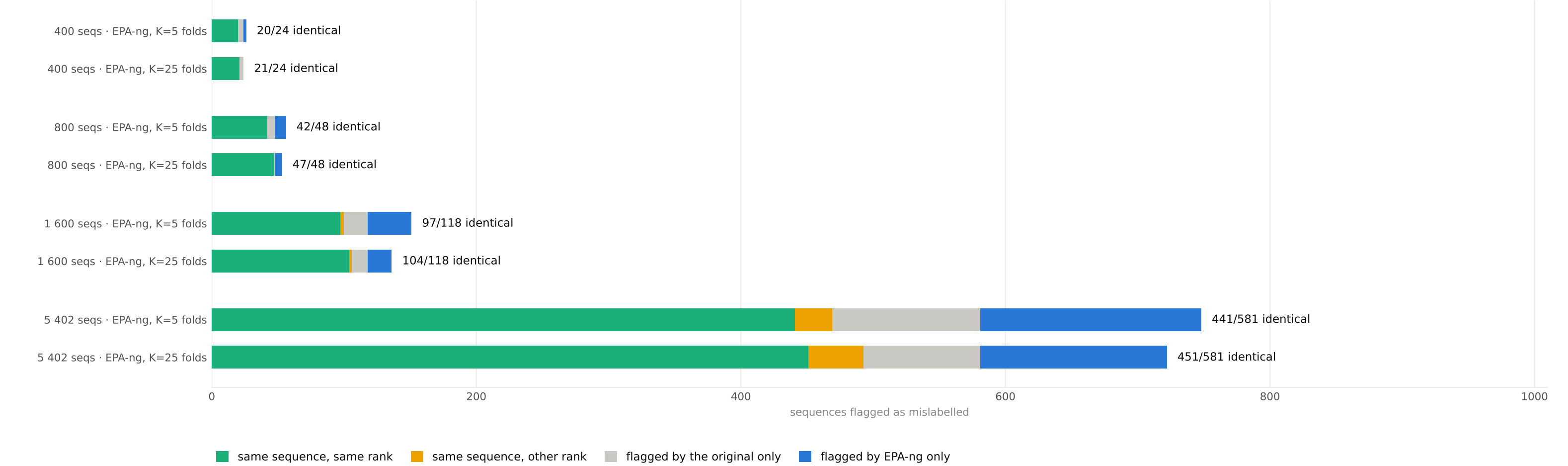
A · Runtime, split by phase

Upstream v0.9.3 is SATIVA as its authors ship it, python 3 included. The EPA-ng version is that same code with the placement engine swapped: the decision-logic files are byte-identical, the diff is two hunks in sativa.py, one in config.py and one new module. The badge over each panel says what SATIVA asks RAxML to do at that size: past 500 taxa GTRCAT instead of GTRGAMMA, past 1 000 taxa a thorough insertion on a fraction of the branches only. Those shortcuts barely change its output (119 flags against 118 at n=1 600) but they do make the orange bar a cheaper computation from n=800 on, which is why the ratio stops growing. The EPA-ng version keeps its own reference tree on GTRGAMMA at every size, so that EPA-ng reads a fitted shape parameter.



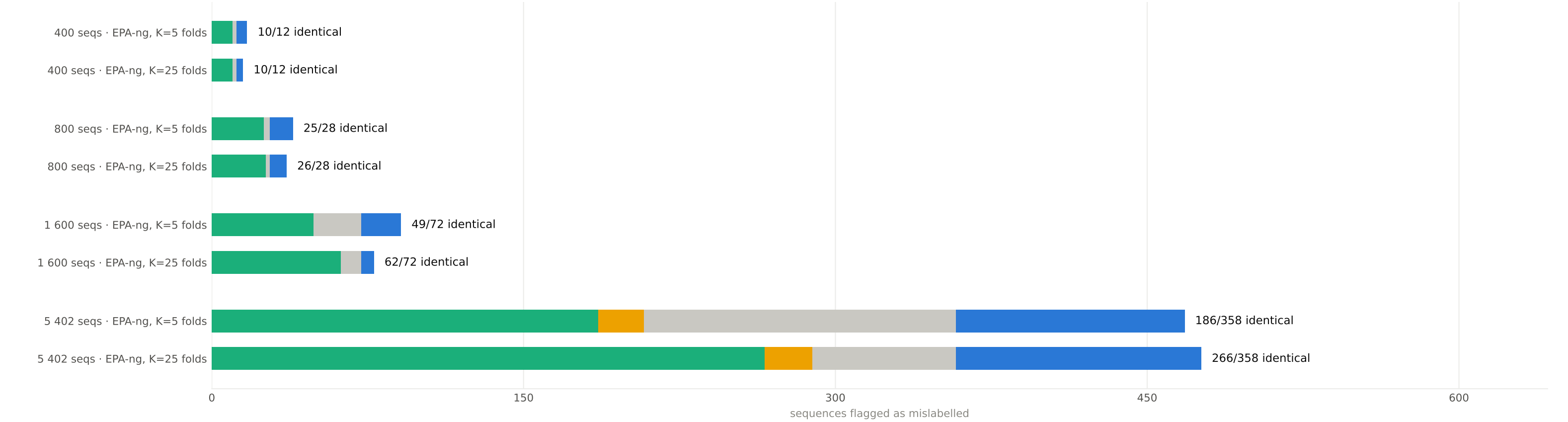
B · Per-sequence agreement with unmodified SATIVA

Each row: the EPA-ng version against unmodified SATIVA v0.9.3, sequence by sequence, cutoff 0.4 on both sides. K is the number of folds the leave-one-out is split into, the only approximation in the modification and the knob that closes most of the gap. At a fixed seed the replicates give identical calls, so none of this is run-to-run noise.



C · The same, keeping only calls above 0.9 confidence

The same comparison with both sides filtered at confidence ≥ 0.9 . It does not make the two agree: with K=25, recall is 0.93 at n=800, 0.86 at n=1 600, 0.81 at n=5 402. And confidence predicts reproducibility only at the bottom of its range: calls between 0.40 and 0.50 are reproduced 0.69 of the time, against about 0.8 above, with a call at 0.95 no safer than one at 0.6.



Medians of the replicates (spread in summary.tsv). Every run works on local disk: on the project's network filesystem the I/O dominates and adds several-fold noise. Panel A starts at 400 sequences and panel B follows it. Below about 400, SATIVA's whole-second phase timer cannot resolve the split, and the smaller sizes stay in summary.tsv and concordance.tsv. In panel B the k-fold leave-one-out is an approximation of the per-sequence one, so it drops and adds a few flags.